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Parameter identification method of the semi-coupled fracture model for 6061 aluminium alloy sheet based on machine learning assistance

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ABSTRACT

The ability to describe the fracture behaviour of materials and the difficulty of parameter calibration are important factors for evaluating the ductile fracture criterion. To find a ductile fracture criterion that can balance both these factors, in this study, the fracture behaviour of 6061 aluminium alloy was characterized based on a semi-coupled damage mechanics framework, and a parameter identification method based on machine learning assistance was proposed. Using this method, the calibration method of the damage initiation model was improved to fully consider the non-linearity of the loading path. Moreover, the calibration process of the fracture and damage evolution models was replaced by the neural network model. The results indicated that the semi-coupled fracture model can accurately predict the damage initiation, accumulation, and fracture of the 6061 aluminium alloy during the forming process. The parameter identification method based on machine learning assistance can circumvent the process of extracting the stress state evolution from the numerical models of calibrated specimens; a set of reliable parameters can be extracted from the force–displacement response using only one specimen. The combination of the semi-coupled fracture model and parameter identification method provides a new approach to balance the difficulty of parameter calibration and the description ability of the ductile fracture criterion.

1. Introduction

With the development of lightweight automobiles, aluminium alloy sheets have been widely used in manufacturing to reduce body weight (Haghshenas and Gerlich, 2018; Hirsch, 2014; Lee et al., 2018). However, the plasticity of aluminium alloys at room temperature is limited, and cracks are prone to occur during the forming process, which causes difficulties in the forming process and limits their use. Therefore, it is important to extensively study the mechanical behaviour and particularly the fracture behaviour of aluminium sheets (Kubik et al., 2018; Khadyko et al., 2019; Ji et al., 2020).

Fracture models can be divided into coupled and uncoupled models based on the interaction between the damage and plastic behaviour. The uncoupled model ignores the influence of damage on plastic behaviour, plastic development still follows classical theory (Han et al., 2021; Lou and Yoon, 2019; Lou et al., 2012). Typical uncoupled models include the Mohr–Coulomb (Bai and Wierzbicki, 2008), Rice and Tracey (Rice and Tracey, 1969), and Johnson–Cook criteria (Johnson and Cook, 1985). The coupled model considers that the plastic development process is accompanied by the evolution of damage, and the accumulation of damage leads to a decline in material performance, which eventually induces fracture occurrences (Reddi et al., 2019; Baltic et al., 2020; Baltic et al., 2020). Typical coupled models include the Gurson–Tvergaard–Needleman (GTN) (Li and Cui, 2020) and Lemaitre damage models (Bergo et al., 2021). The uncoupled fracture model has the advantages of simple structure and easy parameter identification; however, the damage evolution of materials cannot be described, and the prediction accuracy of the model is lower when the degree of damage is higher. The coupled fracture model can describe the damage evolution process more accurately; however, the difficulty of parameter identification limits the practical application of these models.

By integrating the advantages and disadvantages of coupled and uncoupled models, some studies (Pütz et al., 2020; Shen et al., 2020) have embedded the uncoupled fracture criterion into the damage

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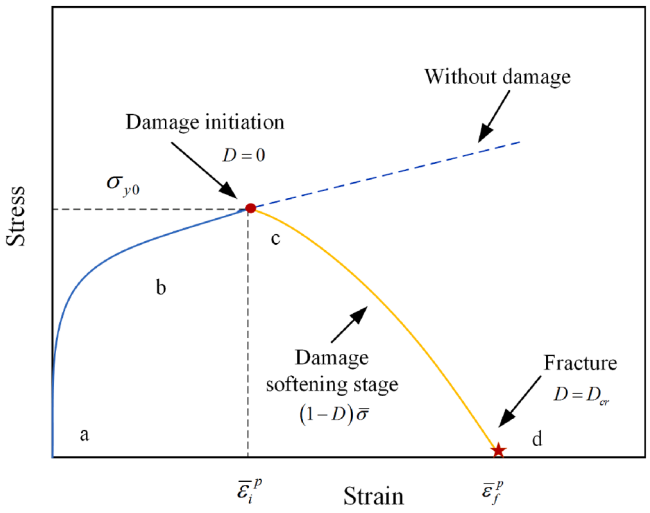


Fig. 1. Stress–strain curve exhibiting damage development.

mechanics framework and proposed a semi-coupled fracture model. Lian et al. (2013), Lian et al. (2015) modified the fracture strain in the Bai–Wierzbicki fracture model as the initial damage strain, coupled the accumulative damage variable to the hardening equation to describe the damage softening effect, and established a concise semi-coupled ductile fracture criterion. Wu et al. (2017) proposed the concept of damage initiation and fracture indicators considering the non-proportional loading path and improved the Bai–Wierzbicki model by considering the influence of stress triaxiality and Lode angle parameters. Liu et al. (2020) considered the effects of the stress state, temperature, and strain rate based on a semi-coupled fracture model, which improved the effectiveness and flexibility of the model. The semi-coupled fracture model based on the uncoupled fracture criterion not only considers the effect of damage, but its parameter calibration is simpler than that of the coupled model. Therefore, this study attempts to construct a ductile fracture model with strong descriptive abilities and easy identification of parameters based on a semi-coupled damage mechanics framework.

Regardless of the form of the fracture model, the calibration of the fracture locus depends on the extraction of the stress state. The process of directly measuring the stress state of a specimen through experiments is complicated. Therefore, the experimental–numerical hybrid method is typically used for calibration. However, this method requires a finite element simulation of each specimen involved in the calibration to extract the evolution process of the stress state and strain field. The calibration process is challenging, and the calibration accuracy depends on the extracted stress state and strain field. Dunand and Mohr (2010) demonstrated that the selection of extraction elements and time discretization errors has a certain impact on the stress state and strain field. For the semi-coupled fracture model, this effect is further amplified owing to the interference of damage; moreover, some parameter calibrations involve the inversion method, which further increases the difficulty of calibration. Recent studies have indicated that machine learning has significant potential for parameter identification (Sun et al., 2021; Yao et al., 2021; Guo et al., 2021). Shikalgar et al. (2020) trained a neural network with force–displacement data as the input, and GTN parameters as the output, to determine the parameters of the GTN model. Baltic et al. (2021) introduced the force–displacement response and strain field of a shear specimen obtained by a digital image correlation method into an artificial neural network to identify the parameters that predict the real mechanical response. Pandya et al. (2020) expressed the stress–strain response through a neural network model, which was trained only on the basis of data, without making assumptions about the existence of the flow or hardening laws. Based on these works, this study attempts to calibrate the semi-coupling model using a neural network to simplify the calibration process and reduce calibration difficulty.

To balance the description ability and parameter calibration difficulty of the ductile fracture criterion, this study embedded the three-dimensional damage and fracture loci based on the uncoupled fracture criterion of Lou–Huh into the semi-coupled damage mechanics framework; combined with the modified Bai–Wierzbicki model, the fracture behaviour of 6061 aluminium alloy was studied. A parameter identification method based on machine learning assistance was proposed, which replaces the calibration process of the damage evolution and fracture models using the neural network model. The combination of the semi-coupled fracture model and parameter calibration method simplifies the process of parameter calibration and reduces the difficulty of calibration while ensuring description accuracy.

2. Semi-coupled fracture model

A stress–strain curve containing the damage development is shown in Fig. 1. If the effect of damage on the plastic model is not considered, the material response develops along the dotted blue line. However, damage accumulation caused by the nucleation, growth, and coalescence of voids influences the mechanical properties, and the actual material response develops along the solid yellow line. Based on the aforementioned damage mechanism, Lian et al. (2013) embedded the uncoupled fracture criterion into the damage mechanics framework and proposed a semi-coupled fracture model. The model considers that the hardening of materials is not affected by damage until the strain reaches the damage initiation strain, and the damage variable is activated only in the damage-softening stage, as shown in Fig. 1. Points a–c shows the undamaged stage; point c indicates the onset of damage initiation, point d indicates the fracture, and points c–d indicates the damage development stage. The model consists of three parts:

1. A plastic model for describing the undamaged stage;
2. A damage initiation model for judging damage initiation;
3. A damage evolution model for describing the softening stage between damage initiation and final fracture.

2.1. Modelling of plasticity

The concepts of stress triaxiality η and Lode angle parameter $\bar{\theta}$ are introduced to characterise the three-dimensional stress state. Any three-dimensional stress state can be represented by these parameters, and the expressions are as follows:

$$\eta = \frac{I_1}{3\sqrt{3J_2}} \tag{1}$$

$$\bar{\theta} = 1 - \frac{2}{\pi}\arccos\left(\frac{27J_3^2}{2\bar{\sigma}^3}\right) \tag{2}$$

where I_1 is the first invariant of the stress tensor, J_2 is the second invariant of the stress deviator, J_3 is the third invariant of the stress deviator, and $\bar{\sigma}$ is the equivalent stress.

Owing to the effects of the stress state on plasticity, the true stress–strain curves obtained by uniaxial tensile test cannot accurately describe the yield behaviour of different specimens. Bai and Wierzbicki (Bai and Wierzbicki, 2008) proposed the MBW model by considering the effects of stress triaxiality and the Lode angle parameter on the yield behaviour:

$$\Phi = \bar{\sigma}_{eq} - (1-D)\cdot\sigma_y(\bar{\varepsilon}^p, \eta, \bar{\theta}) \tag{3}$$

$$\sigma_y(\bar{\varepsilon}^p, \eta, \bar{\theta}) = \sigma_y(\bar{\varepsilon}^p)[1 - c_\eta(\eta - \eta_0)]\left[c_\theta^s + \left(c_\theta^{ax} - c_\theta^s\right)\cdot\left(\gamma - \frac{\gamma^{m+1}}{m+1}\right)\right] \tag{4}$$

Table 1
Chemical composition of AA6061 sheet (weight %).

Si	Fe	Cu	Mg	Zn	Ti	Cr	Mn	Al
0.5560	0.3383	0.2524	1.0050	0.0102	0.0317	0.1983	0.0488	Bal

$$\gamma = \frac{\sqrt{3}}{2-\sqrt{3}}\left[sec\left(\frac{\overline{\theta}\cdot\pi}{6}\right)-1\right] \tag{5}$$

$$c_\theta^{ax} = \begin{cases} c_\theta^t, & \overline{\theta} \geq 0 \\ c_\theta^c, & \overline{\theta} < 0 \end{cases} \tag{6}$$

where $\overline{\sigma}_{eq}$ is the equivalent stress, D is the scalar variable used to quantify the damage effects, and $\sigma_y(\overline{\varepsilon}^p, \eta, \overline{\theta})$ is the yield stress determined by the equivalent plastic strain $\overline{\varepsilon}^p$, stress triaxiality η, and Lode angle parameter $\overline{\theta}$. Φ is the yield condition. c_η is a parameter describing the effect of hydrostatic pressure, and η_0 is the value of the stress triaxiality corresponding to the test condition used to determine the stress–strain curve. c_θ^s, c_θ^t, and c_θ^c are the normalized strength under the shear, tension, and compression states, respectively, and m denotes a material parameter with positive integral values that describes the Lode angle sensitivity. γ is another stress state parameter associated with the Lode angle parameter. $\sigma_y(\overline{\varepsilon}^p)$ is a reference flow curve determined using a quasi-static tensile test at room temperature.

The Swift isotropic hardening model was employed to simulate the stress–strain relationship of undamaged materials based on the assumptions of isotropy and plastic incompressibility, as follows:

$$\sigma_y(\overline{\varepsilon}^p) = A(\overline{\varepsilon}^p + B)^n \tag{7}$$

2.2. Damage initiation and fracture models

During plastic deformation, the ideal stress state is difficult to maintain and is an evolving process. The average values of the stress triaxiality η_{avg} and Lode angle parameter $\overline{\theta}_{avg}$ are typically used to characterise the overall stress state:

$$\eta_{avg} = \frac{1}{\overline{\varepsilon}^p}\int_0^{\overline{\varepsilon}^p} \eta(\overline{\varepsilon}^p)d\overline{\varepsilon}^p \tag{8}$$

$$\overline{\theta}_{avg} = \frac{1}{\overline{\varepsilon}^p}\int_0^{\overline{\varepsilon}^p} \overline{\theta}(\overline{\varepsilon}^p)d\overline{\varepsilon}^p \tag{9}$$

The damage initiation model is the criterion for judging the beginning of the damage evolution. Constructing a reasonable damage initiation model is key to accurately describing the subsequent damage softening phenomenon. In this study, the uncoupled fracture criterion proposed by Lou et al. (2012) was used as the equation of the damage initiation locus (DIL) to describe the relationship between the critical equivalent plastic strain and the stress state at the damage initiation moment. Based on a previous study (Wu et al., 2017), a cut-off value of the stress triaxiality exists at $\eta_c = -1/3$, below which the material is in the compression state, and the initiation and evolution of damage cannot be triggered. Therefore, when stress triaxiality is lower than η_c, the value of $\overline{\varepsilon}_{ddi}$ is set to infinity:

$$\overline{\varepsilon}_{ddi}^p(\eta,\overline{\theta}) = \begin{cases} +\infty & \eta \leq \eta_c \\ \dfrac{b_3}{\left(\dfrac{2}{\sqrt{\overline{\theta}^2+3}}\right)^{b_1}\left(\dfrac{\langle 1+3\eta\rangle}{2}\right)^{b_2}} & \eta > \eta_c \end{cases} \tag{10}$$

Considering the effects of stress state evolution, the damage initiation indicator I_{dd} is defined as the integral form of the equivalent plastic strain to describe the onset of damage, as follows:

$$I_{dd} = \int_0^{\overline{\varepsilon}^p} \frac{d\overline{\varepsilon}^p}{\overline{\varepsilon}_{ddi}^p(\eta,\overline{\theta})} \tag{11}$$

When I_{dd} reaches one, it indicates the onset of damage. The values of the equivalent plastic strain and stress at the moment of damage initiation ($I_{dd} = 1$) are defined as $\overline{\varepsilon}_{ddi}^{p,c}$ and $\overline{\sigma}_{ddi}^c$, respectively:

$$\overline{\varepsilon}_{ddi}^{p,c} = \overline{\varepsilon}^p \quad (I_{dd} = 1) \tag{12}$$

$$\overline{\sigma}_{ddi}^c = \overline{\sigma} \quad (I_{dd} = 1) \tag{13}$$

Similarly, the equation of the ductile fracture locus (DFL) is used to describe the relationship between the critical equivalent plastic strain and the stress state at the fracture moment. When the stress triaxiality is lower than η_c, the value of $\overline{\varepsilon}_{df}^p$ is set to infinity:

$$\overline{\varepsilon}_{df}^p(\eta,\overline{\theta}) = \begin{cases} +\infty & \eta \leq \eta_c \\ \dfrac{c_3}{\left(\dfrac{2}{\sqrt{\overline{\theta}^2+3}}\right)^{c_1}\left(\dfrac{\langle 1+3\eta\rangle}{2}\right)^{c_2}} & \eta > \eta_c \end{cases} \tag{14}$$

The fracture indicator I_{df} is used to describe the ductile fracture of materials, and is expressed as follows:

$$I_{df} = \int_{\overline{\varepsilon}_{ddi}^{p,c}}^{\overline{\varepsilon}^p} \frac{d\overline{\varepsilon}^p}{\overline{\varepsilon}_{df}^p(\eta,\overline{\theta}) - \overline{\varepsilon}_{ddi}^p(\eta,\overline{\theta})} \tag{15}$$

When I_{df} reaches one, it indicates the occurrence of ductile fracture.

2.3. Damage evolution model

When the material reaches the damage initiation condition (after the damage indicator I_{dd} reaches one), damage evolution is controlled by the energy dissipation theory. When the fracture indicator I_{df} reaches one, the damage variable D does not necessarily reach one. Therefore, a critical value of the damage variable D_{cr} exists, when D reaches this critical value, the material point fails:

$$D_{cr} = \frac{\overline{\sigma}_{ddi}^c}{G_f}\left(\overline{\varepsilon}_{df}^p - \overline{\varepsilon}_{ddi}^p\right) \tag{16}$$

where G_f denotes the energy dissipated at a material point between the damage initiation and fracture.

The damage evolution process is expressed by a linear expression, and the damage variable can be defined as:

$$D = \begin{cases} 0 & I_{dd} < 1 \\ D_{cr}\cdot I_{df} & I_{dd} \geq 1 \wedge I_{df} < 1 \\ 1 & I_{dd} \geq 1 \wedge I_{df} \geq 1 \end{cases} \tag{17}$$

When the damage initiation indicator I_{dd} is less than one, the plastic strain accumulates and no damage occurs. When the damage initiation indicator I_{dd} reaches one, the damage begins to develop, and the damage evolution follows the energy law. When the fracture indicator I_{df} reaches one, the carrying capacity of the material is completely lost, leading to the occurrence of a final fracture.

3. Experiments

In this study, a 6061 aluminium alloy of thickness 2 mm was selected; its chemical composition is listed in Table 1. All the specimens were cut along the rolling direction from the same sheet by electrical

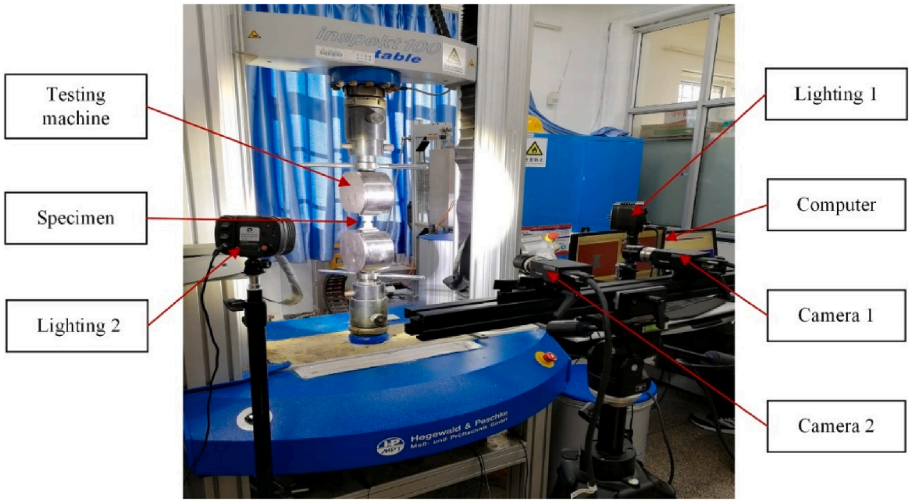


Fig. 2. Experimental device based on DIC system.

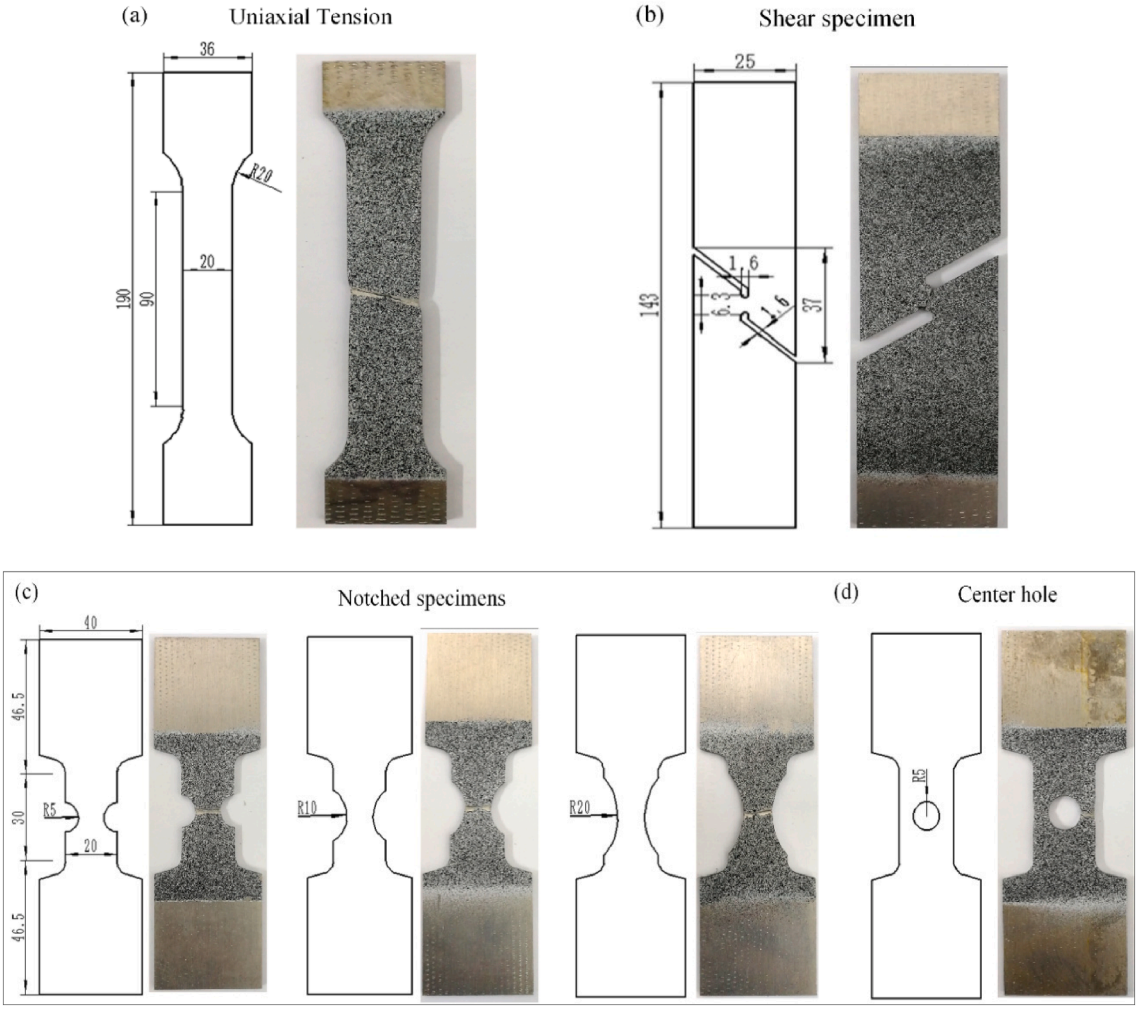


Fig. 3. Shape and size of applied specimens (mm): a) UT b) SH c) NT d) CH.

discharge machining. The experimental scheme consisted of two parts:

1. Uniaxial tensile test to obtain the plastic-hardening behaviour under uniaxial tension;
2. Fracture tests with various specimens to characterize the fracture behaviour of the material in a wide range of stress states.

All the specimens were stretched using a universal material-testing machine. To obtain the strain field of the specimen during the

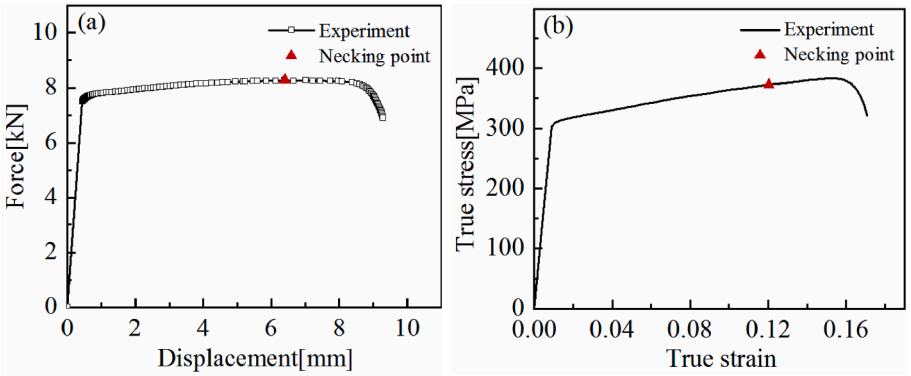


Fig. 4. Uniaxial tensile specimen test: a) Experimental force–displacement curve b) True stress–strain curve.

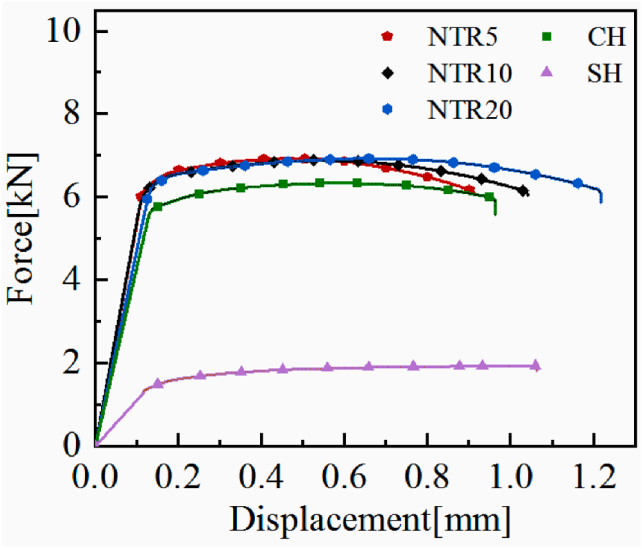


Fig. 5. Force–displacement curves of the fracture test.

deformation process, a digital image correlation (DIC) system was used to monitor the entire experimental process. As shown in Fig. 2, the DIC measurement system consists of a binocular vision acquisition system, a lighting system, and an information acquisition system. Uniform black and white speckles of diameter approximately 0.1 mm were sprayed onto the surface of the specimen in advance, and two cameras (MER-030–120-GM/C-P) were used to record images with a resolution of 2048 × 2048 pixels at a speed of 5 images per second during the experiment. The collected images were analysed using vic-3D software to determine the surface displacement and strain fields. The subset size was set to 17 pixels and the step size was set to 5 pixels.

3.1. Uniaxial tensile test

The uniaxial tensile specimen was designed to test the mechanical property of the material, and its size is shown in Fig. 3a. The specimens were stretched using a universal material testing machine until fracture, and the loading speed was set to 0.5 mm/min. After the experiment, the force–displacement curve of the gauge length (50 mm) was obtained, as shown in Fig. 4a. The true stress and strain are calculated as follows:

$$\varepsilon = \ln\left(1 + \frac{\Delta l}{l_0}\right) \tag{18}$$

$$\sigma = \frac{F}{A_0}\left(1 + \frac{\Delta l}{l_0}\right) \tag{19}$$

where ε and σ denote the true strain and stress, respectively, l_0 is the initial gauge length, Δl is the instantaneous elongation of the gauge length during deformation, A_0 is the initial cross-sectional area of the specimen, and F is the load.

The true stress–strain curve is shown in Fig. 4b. Because the aforementioned formulas are only applicable when the specimen does not exhibit necking, only the part of the true stress–strain curve prior to the necking point is used in subsequent calibrations. The onset of necking is defined as the highest point on the load curve.

3.2. Fracture test

To study the fracture behaviour of the material in a wide range of stress states, five specimens were designed: shear specimen (SH), notched specimens with radii of 5 mm (NTR5), 10 mm (NTR10), and 20 mm (NTR20), and a centre hole specimen (CH). The specimen sizes are shown in Fig. 3. The tensile speed of the universal material testing machine was set to 0.5 mm/min. After the experiment, the force–displacement curve of each specimen was extracted, as shown in Fig. 5.

4. Identification method based on machine learning assistance

The experimental–numerical hybrid method relies on the extraction of the stress state and strain fields. The selection of the critical element and the time discretization error influence the evolution of the equivalent plastic strain and stress state, which affects the calibration of the parameters. To overcome these problems, this study attempts to replace the traditional experimental–numerical hybrid method with machine learning. The number of parameters that need to be calibrated in the semi-coupled model is large, which presents a challenge for the neural network, and the number of parameters must be appropriately reduced. The modified plastic (Eq. (4)), and damage initiation models (Eq. (10)) are not affected by the damage evolution, and the stress state extracted from the non-damage plastic model are consistent with the semi-coupled model. Therefore, this study employed the experimental–numerical hybrid method to calibrate these parameters, and the calibration process of the fracture and damage evolution models were replaced by the neural network. The parameters to be identified by the neural network are (c_1, c_2, c_3, G_f). In theory, as more types of specimens are considered, the mapping capability of machine learning becomes stronger; however, the requirement for neural network is higher for more types of specimens, and a more numerical simulation is needed to create the database. Considering the performance of the neural network and cost of creating the database, only one specimen was used for machine learning in this study. The calibration process is illustrated in Fig. 6.

1. A series of tests covering a broad range of stress states were performed, the force–displacement response and strain field during the entire deformation until fracture were recorded.

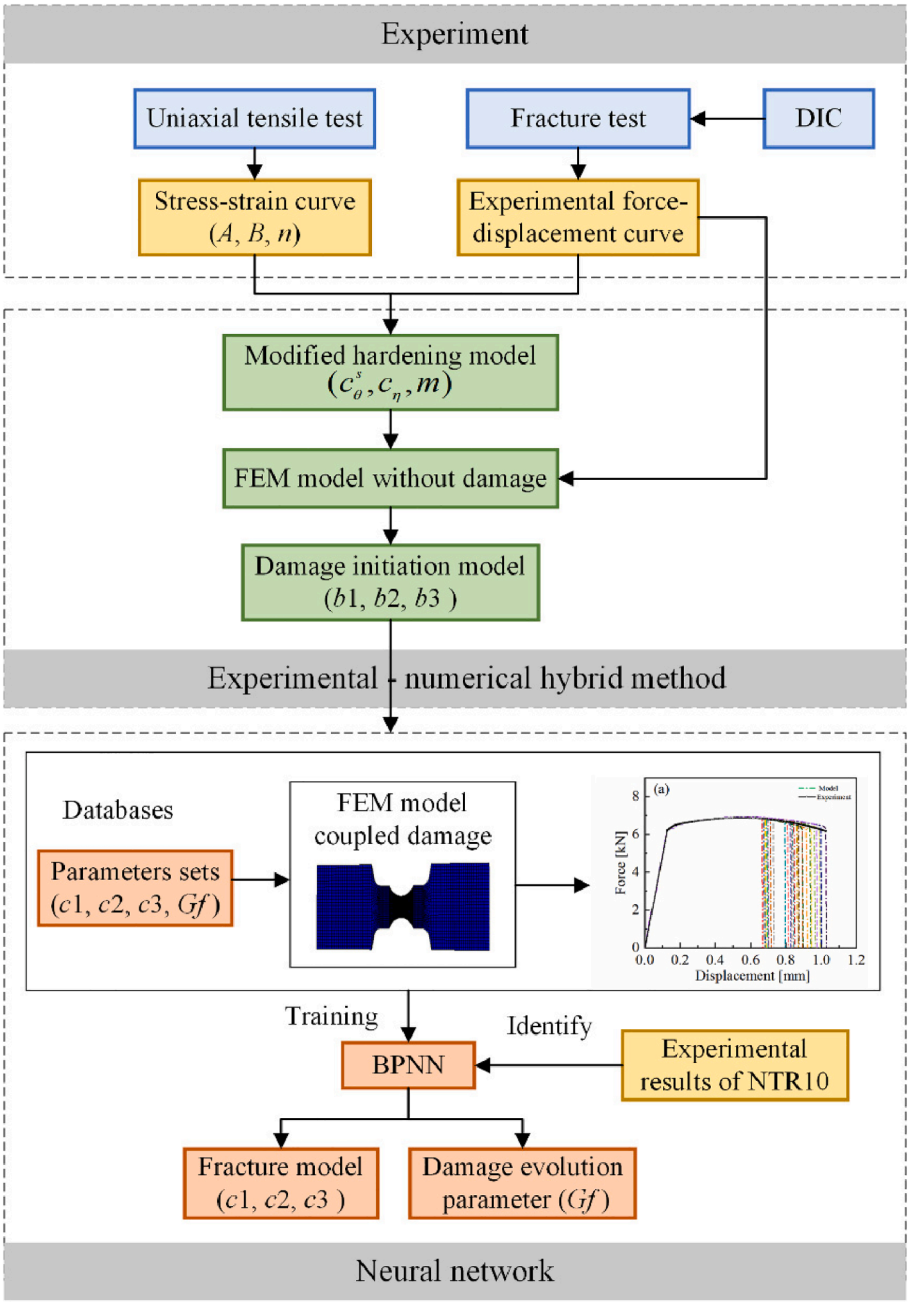


Fig. 6. Parameter identification process based on machine learning assistance.

2. The parameters of the Swift hardening model were calibrated based on the data of the uniaxial tensile test.
3. The finite element simulation of all specimens involved in calibration was performed based on the Swift hardening model, and parameters related to the stress state in the modified model were calibrated by comparing the experimental and simulated force–displacement responses in the no-damage plastic stage.
4. The finite element simulation of all specimens involved in calibration were performed based on the modified hardening model. The damage initiation moment was determined by comparing the experimental and simulated force–displacement responses. The damage initiation strain and stress states evolution processes were extracted from the simulation results to calibrate the damage initiation model.
5. NTR10 was selected as the training specimen, and different parameter sets (c_1, c_2, c_3, G_f) were selected for embedding into the semi-coupled model for finite element simulation. The force–displacement curve was extracted to construct the training database of the neural network, and the neural network was trained based on the database.
6. The experimental force–displacement data of NTR10 were used as the input of the neural network to predict the parameter set.

4.1. Calibration of hardening model

The Swift isotropic hardening model (Eq. (7)) was used to fit the flow curves of the material. The values of A, B, and n in the model were determined using the least squares method. The promising fitting result is shown in Fig. 7. The calibration results are listed in Table 2.

After the Swift model was determined, the parameters representing the influence of the stress state in the MBW model must be calibrated (Eq. (4)). Because the stress–strain curve was determined by the uniaxial

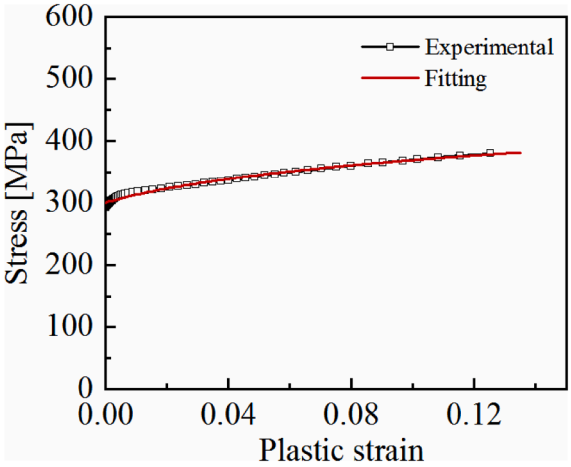


Fig. 7. Fitting results of Swift model.

Table 2
Parameter calibration results of the semi-coupled model.

Model name	Parameters calibration results				
Plastic model	A [MPa]	B	n		
	488.01	0.029	0.1365		
Modified hardening model	c_η	c_θ^t	c_θ^s	c_θ^c	m
	0.05	1.0	0.95	1.0	4
Damage initiation	b_1	b_2	b_3		
	3.434	1.064	0.111		
Damage evolution and fracture	G_f[mJ/mm^3]	c_1	c_2	c_3	
BP neural network	458.38	0.9359	0.8689	0.4884	
Experimental–simulation	551.23	3.0327	0.7682	0.5926	

tensile test, the value of η_0 was 1/3, and the value of c_θ^t was one. In this study, the stress state was mainly concentrated in the range of the Lode angle parameter greater than zero. Therefore, the value of c_θ^c was set to one. The parameters that need to be calibrated were c_θ^s, c_η, and m, determined by the finite element inversion method, to ensure that the force–displacement curves of each specimen between the simulation and experiment were the most consistent. The calibration results are listed in Table 2, and the experimental and numerical force–displacement responses are compared in Fig. 8.

4.2. Calibration of damage initiation model

A shear specimen and three notched specimens were used for calibration. All the experiments were simulated using a modified hardening model. The boundary conditions of the model are the same as those of the experiment; one end is fixed, and the other end is subjected to displacement constraints. To avoid the severe impact of the mesh size, a consistent element type and size were employed for all models, with fine mesh in the deformation area and coarse mesh in the remaining area. The element type was set as a three-dimensional eight-node solid element with reduced integration (C3D8R). The moment of damage occurrence must be identified to determine the damage-initiation strain. Using the method proposed in Ref. (Børvik et al., 2001; Bouchard et al., 2011), the point at which the simulated force–displacement curve based on the non-damage model (Eq. (4)) deviates from the experimental force–displacement curve was regarded as the macroscopic mechanical point of damage initiation. In this study, for each specimen, there is an intersection point between the experimental and simulated force–displacement curves, after this point, the distance between the two curves gradually increases. Therefore, the last intersection point of the experimental and simulated force–displacement curves was taken as the onset of damage initiation. The calibration results for the damage initiation moment of each specimen are shown in Fig. 8.

The integration point with the highest strain at the damage initiation moment was selected as the critical point, and the equivalent plastic strain at the damage initiation moment of this point was extracted as the damage initiation strain. The strain history and the evolution of the stress state from plastic initiation to damage initiation were extracted. The evolution of the stress triaxiality and Lode angle parameter is shown in Fig. 9.

The stress state constantly evolves during plastic deformation, the average stress triaxiality and average Lode angle parameter are typically used to describe the stress state during loading. To reflect the nonlinearity of loading paths, this study proposed a calibration method by evaluating the damage initiation indicator I_{dd} as opposed to the calibration method based on the average stress state. Substituting the extracted three-dimensional damage initiation locus (curve in Fig. 9) into the integral expression of the damage initiation indicator (Eq. (11)), the parameters in the damage initiation model are determined based on the optimization algorithm. The optimization objective is set as:

$$\{(b_1,b_2,b_3)\} = \left\{ (b_1,b_2,b_3) \middle| \sum_{i=1}^{N_s} \left(I_{dd}^i(b_1,b_2,b_3) - 1\right)^2 \rightarrow \min \right\} \tag{20}$$

where I_{dd}^i denotes the damage initiation indicator of the specimens involved in the calibration, and N_s denotes the number of calibration specimens.

The parameter results calibrated using the optimization algorithm are listed in Table 2. The damage initiation surface of the 6061 aluminium alloy and the three-dimensional damage initiation locus are shown in Fig. 10.

4.3. Calibration of damage evolution and fracture models based on the neural network

4.3.1. Establishment of training database

To obtain a neural network model with high prediction accuracy, high-quality training data is vital. In this study, the values of (c_1, c_2, c_3, G_f) were preliminarily calibrated based on the fracture strain extracted by DIC, stress state determined during the calibration of the damage initiation model, and previous studies. The calibrated value of (c_1, c_2, c_3, G_f) was taken as the centre, and each parameter was represented by three levels: low, medium, and high, as shown in Table 3. The selected range of parameters must ensure that the experimental curve is within the spatial domain covered by the training database, as shown in Fig. 11. A three-level full factorial experimental design was used to obtain a collection of parameter sets, and the total number of parameter sets was $3^4 = 81$.

Other parameters in the semi-coupled model still adopted the calibration results listed in Table 2. The values of (c_1, c_2, c_3, G_f) adopted the data in Table 3, and the 81 groups of parameter sets were used for the finite element simulation. Baltic et al. (Baltic et al., 2021) showed that regardless of the specimen chosen, it is possible to identify the real material parameters through machine learning. In this study, NTR10 was selected as the training specimen because the stress triaxiality and Lode angle parameter of NTR10 are in the relative middle position, as shown in Fig. 9, and a wider stress states is easier to obtain at this centre. The boundary conditions of the model were set identical to those in the experiment. C3D8R was used for all the finite element models, and the mesh sizes of all the models remained the same. After the simulation, the corresponding force–displacement curve was extracted, and 10 data points were extracted at equal time intervals from the stage of damage initiation to the fracture, as shown in Fig. 12. The parameter sets and corresponding force–displacement responses constitute the training database of the neural network.

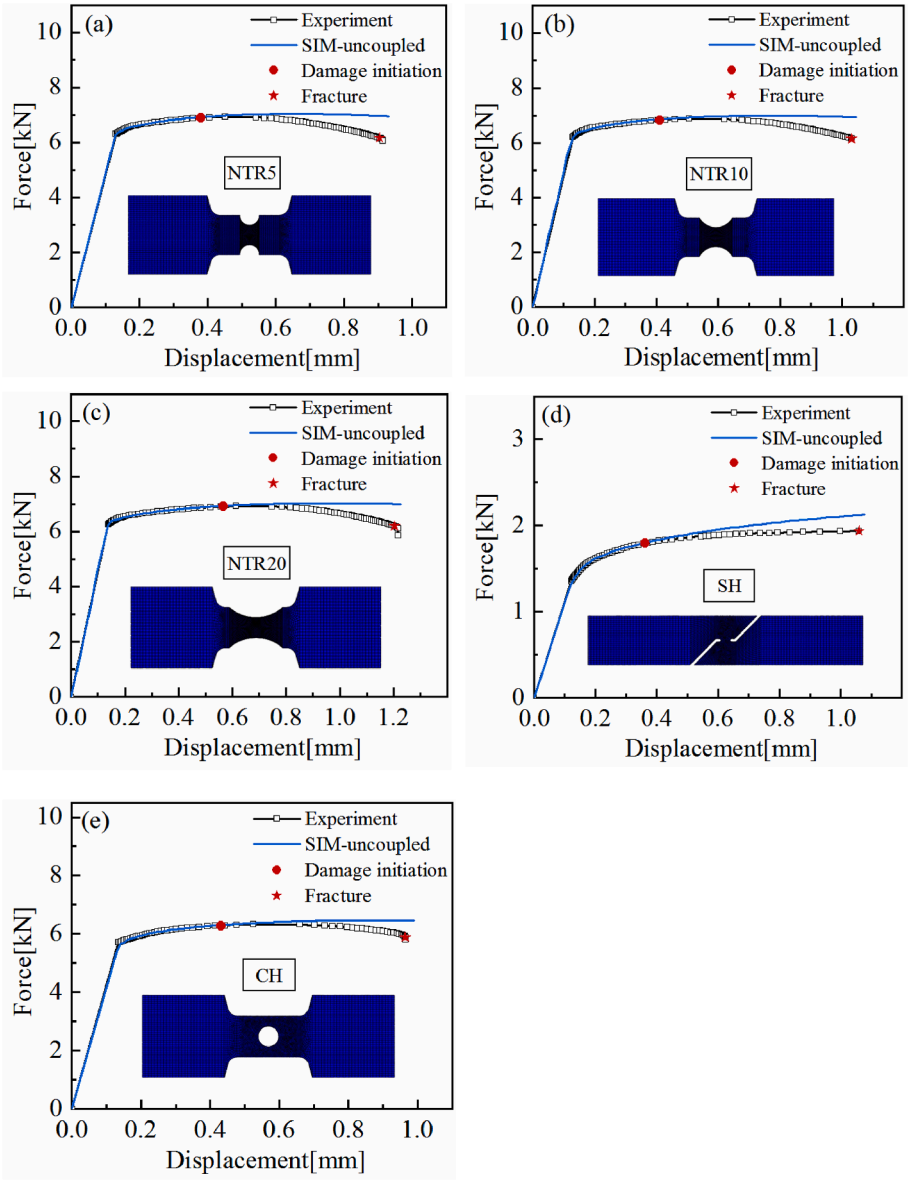


Fig. 8. Comparison of experimental and simulated force–displacement curves: a) NTR5 b) NTR10 c) NTR20 d) SH e) CH.

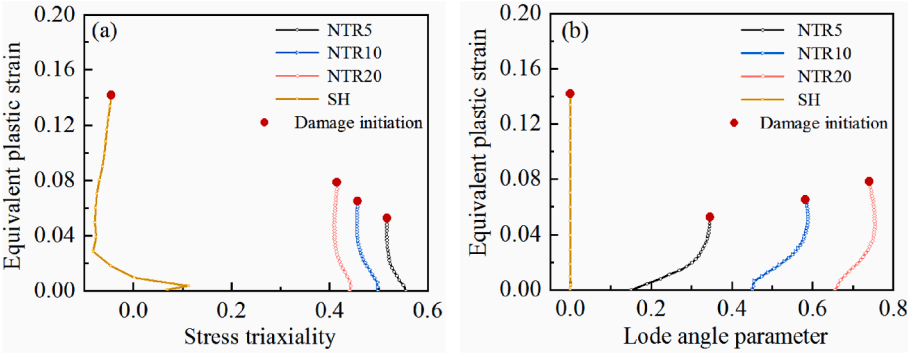


Fig. 9. Loading history of critical point extracted from FEM: a) Stress triaxiality b) Lode angle parameter.

4.3.2. Construction of neural network model

The backpropagation algorithm, which continuously adjusts the weights and thresholds of the network through the backpropagation of errors to minimise the sum of errors, is one of the most widely used supervised machine learning algorithms. The calculation process is as follows:

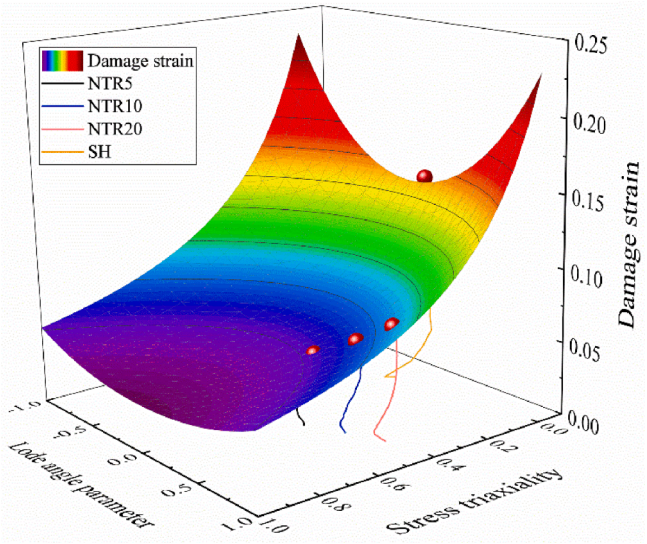


Fig. 10. Damage initiation surface and three-dimensional damage initiation locus.

Table 3
Design of each parameters by three levels.

	c_1	c_2	c_3	G_f
Level 1	0.408	0.469	0.295	400
Level 2	2.048	0.669	0.395	500
Level 3	4.408	0.869	0.495	600

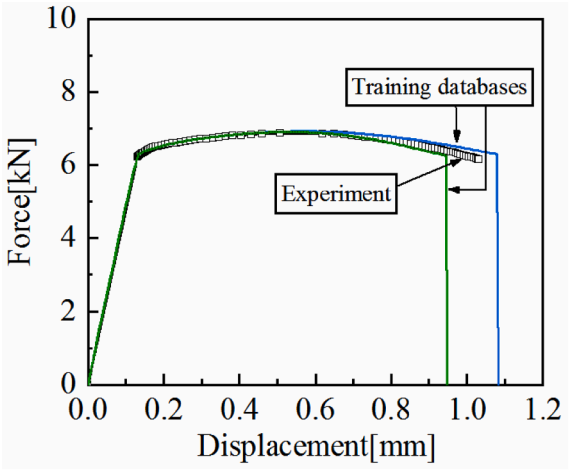


Fig. 11. Relationship between the spatial domain of training database and experimental.

1. All input variables must be normalized in the calculation process and projected to the range [−1, 1] to eliminate the deviation caused by the different scales of each input vector:

$$x^n = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{21}$$

where x and x^n denote the original and normalized data, respectively. $x_{\max}$ and $x_{\min}$ denote the minimum and maximum values of the training data, respectively.

2. The first and subsequent hidden layer are activated, and the weight matrix and deviation vector calculated:

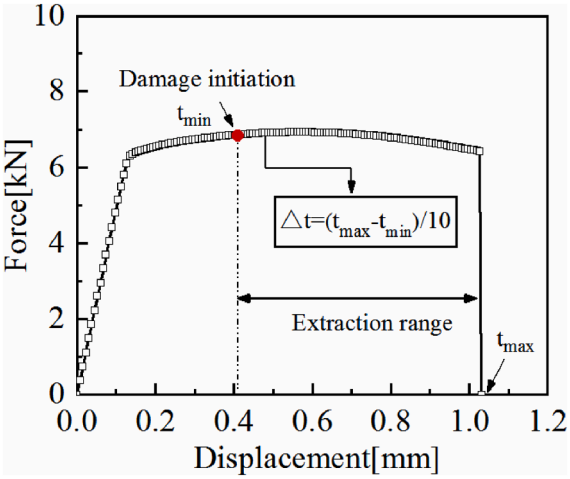


Fig. 12. Passing the training database from FE to BPNN.

$$a_j^{(n)} = h\left(\sum_{i=1}^{n_n} w_{ji}^{(n)} a_i^{(n-1)} + b_j^{(n)}\right) \tag{22}$$

where h denotes the activation function, a_i is the input of the ith node in the input layer, $w^{(n)}$ and $b^{(n)}$ denote the weight matrix and deviation vector, respectively.

3. The output variable is calculated based on the hidden units of the last hidden layer.

$$\bar{y}_j = \sum_{i=1}^{n_n} w_{ji}^{(n_n+1)} a_i^{(n_n)} + b_j^{(n_n+1)} \tag{23}$$

4. The output layer must be projected from [−1, 1] onto the design space by linear scaling.

The parameter identification process based on the BP neural network is shown in Fig. 13. Data points extracted from the force–displacement curve and the corresponding parameter sets (c_1, c_2, c_3, G_f) were set as the input and output of the neural network, respectively. Levenberg–Marquart was set as the training algorithm. 90% of the data in the training database were used for training and learning, and remaining 10% were used to evaluate the predictive ability of the neural network model. The modelling, training, and testing of the neural network were performed using the MATLAB neural network toolbox.

By considering the influence of the neural network structure, models with single, double, and triple hidden layers were established. The number of neurons in all hidden layers kept the same, and the range of numbers was determined as 6 to 15 according to the expressions as follows:

$$n_{\text{hidden}} = \sqrt{n_{\text{input}} + n_{\text{output}}} + z \tag{24}$$

where n_{input} and n_{output} denotes the number of neurons present in the input and output layers, respectively, z denotes an integer ranging from 1 to 10.

To prevent over fitting during training, an L2 regularization penalty was added to the network with a regularization factor of 0.005. The inverse problem of estimating the material parameters in this study was highly non-linear, and different parameter sets may produce the same force–displacement response. Therefore, the predictive ability of the neural network model was not verified by judging the relative error of the predicted parameters, and the final model verification was based on the correct reproduction of the force–displacement response. The pa-

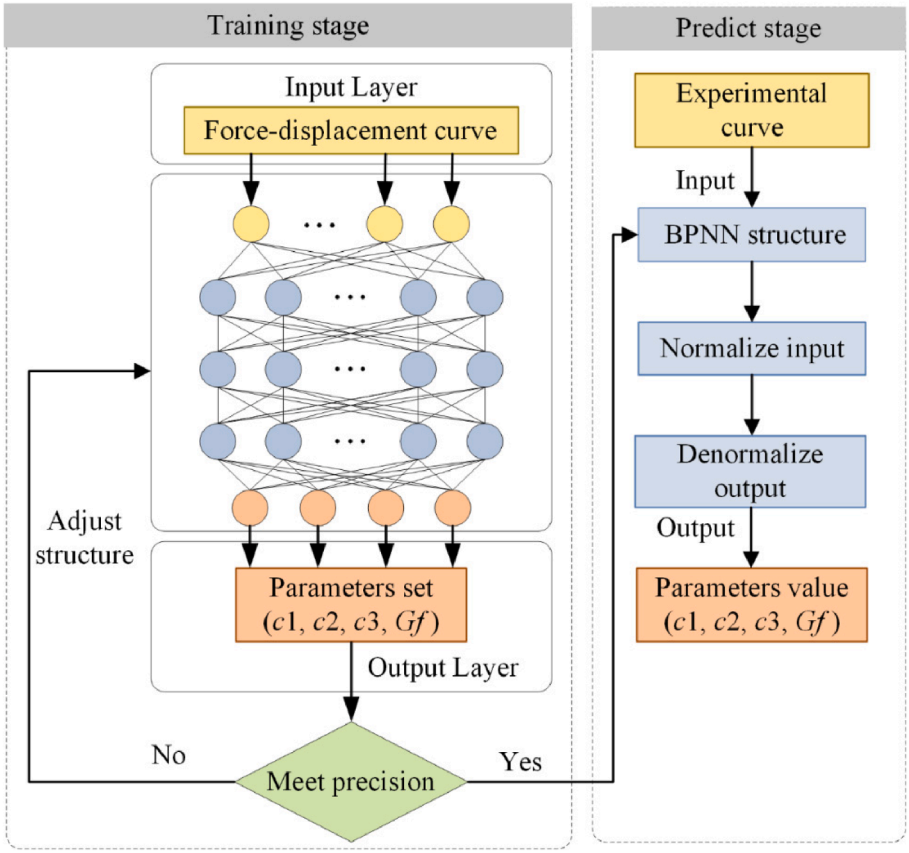


Fig. 13. Logic flow chart of parameter identification based on the neural network.

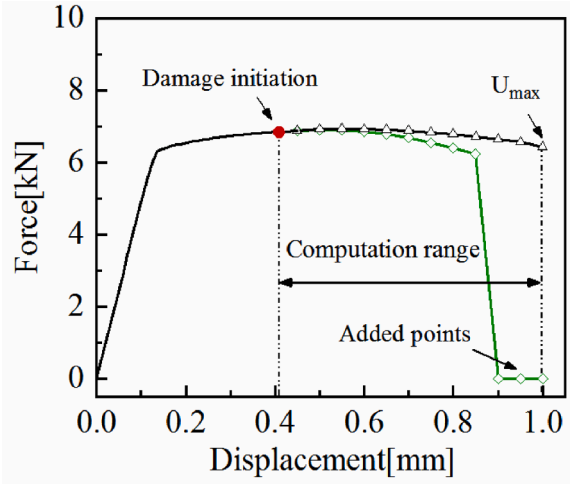


Fig. 14. Error computation method based on force–displacement curve.

rameters predicted by different models were fed back to the finite element simulation, and the corresponding force–displacement curve was extracted. The difference between the extracted force–displacement curve and the input force–displacement curves was taken as the standard to judge the prediction ability. The error was expressed as follows:

$$Error = \sum_{k=1}^{p} \left| F_k^{\text{input}} - F_k^{\text{output}} \right| \tag{25}$$

where p is the number of data points used for computation, F^{input} and F^{output} are the force values of each point on the input force–displacement curve and the force–displacement curve corresponding to the predicted parameter set, respectively.

Considering the prediction ability of the model for the fracture displacement, the displacement range of the error computation was extended to the position with the maximum displacement between the two curves, and the missing force data in the expansion range were replaced by zero values, as shown in Fig. 14.

Eight groups of data that did not participate in training were selected as the test cases to verify the predictive ability of the neural network model. The errors in the models with different structures are listed in Table 4. It is observed that the error decreases as the number of layers are increased from 1 to 3. When the number of neurons increases to 12, the error of the model tends to be stable. The model with the structure of [14–14–14] had the smallest average error, and the error of each test sample was uniform; this implied that the model had good comprehensive predictive and stable generalisation abilities.

4.3.3. *Prediction of neural network*

The neural network model with the structure [14–14–14] was selected through a comparison of the predictive ability. Data points were extracted from the experimental force–displacement curve of NTR10 and set as the input of the neural network to predict the parameters. The results are presented in Table 2.

5. Results and discussion

5.1. *Finite element analysis of machine learning assistance method*

The calibration results of the parameters in the semi-coupled model using the machine learning assistance method are listed in Table 2. Finite element simulations were performed for all the specimens based on the parameter set, including those not involved in the training stage,

Table 4
Errors of test samples corresponding to different neural network structure.

Structure of hidden layer	Error of different test samples [kN]								
	Test 1	Test 2	Test 3	Test 4	Test 5	Test 6	Test 7	Test 8	Average error
6	41.37	34.32	26.93	34.40	27.34	14.11	41.33	20.72	30.07
7	13.82	20.52	51.78	13.75	6.86	72.32	20.67	13.84	26.70
8	31.53	4.08	11.45	31.66	10.88	23.53	38.61	10.91	20.33
9	17.95	17.87	29.73	10.93	4.09	43.46	4.05	11.12	17.40
10	4.07	10.86	48.99	10.93	4.03	36.56	17.88	4.06	17.17
11	10.91	10.85	10.62	10.92	10.90	24.41	31.70	10.95	15.16
12	2.18	15.79	2.62	2.07	22.55	8.70	15.87	29.53	12.42
13	14.86	14.73	1.34	7.93	7.89	1.24	21.77	14.84	10.58
14	8.04	1.20	20.62	1.04	7.92	20.48	7.95	8.12	9.42
15	7.91	7.84	20.20	7.73	14.75	14.17	7.79	7.92	11.04
6–6	8.02	1.05	39.95	7.92	21.69	60.25	14.86	28.74	22.81
7–7	8.06	14.86	27.34	21.75	21.61	14.99	28.64	28.53	20.72
8–8	14.97	8.09	33.30	7.95	1.03	53.42	14.86	1.15	16.85
9–9	8.48	7.98	14.23	14.82	14.74	34.06	14.89	8.10	14.60
10–10	1.69	8.03	20.31	7.95	7.89	34.00	14.85	8.06	12.78
11–11	1.64	14.62	1.76	7.97	14.71	40.2	14.85	7.94	12.90
12–12	1.67	1.17	20.43	1.05	7.88	6.31	7.31	8.31	9.31
13–13	6.99	0.19	13.48	13.84	6.94	7.39	20.78	0.20	8.73
14–14	0.17	6.89	7.07	6.95	6.92	0.65	6.94	13.85	6.18
15–15	0.15	0.18	0.60	6.96	6.94	33.22	6.93	6.94	7.74
6–6–6	20.62	6.89	25.66	20.70	6.92	45.75	13.84	27.62	21.00
7–7–7	13.80	0.12	7.62	27.59	7.13	20.21	34.54	14.19	15.65
8–8–8	13.94	27.34	13.49	6.98	6.89	7.14	13.84	6.95	12.07
9–9–9	13.75	6.85	25.74	0.02	0.07	39.25	13.85	7.00	13.32
10–10–10	0.15	0.10	26.02	0.03	6.93	39.26	6.94	7.07	10.81
11–11–11	20.88	13.79	20.51	6.97	6.91	6.73	6.94	0.22	10.37
12–12–12	0.21	6.83	25.84	6.97	0.09	19.47	0.03	7.06	8.31
13–13–13	13.90	13.66	1.07	6.98	6.88	13.83	6.95	0.09	7.92
14–14–14	0.13	6.92	7.11	0.03	6.88	0.17	6.95	7.05	4.41
15–15–15	0.03	6.84	0.45	6.96	6.93	6.88	13.88	13.83	6.97

and a comparison of the experimental and simulated force–displacement is shown in Fig. 15. The experimental and simulated force–displacement curves of all the specimens exhibited good consistency before damage initiation, indicating that the modified hardening model considering the stress state had a higher description accuracy. After damage initiation, the material exhibited a significant damage-induced softening phenomenon, and the simulation results based on the non-damage model (blue curve) were significantly higher than the experimental results. The semi-coupled fracture model accurately describes the softening phenomenon of the material (yellow curve), which is consistent with the experimental results. The predicted results of the fracture displacement are close to the experimental results, which indicates that the damage evolution and fracture models predicted by the neural network have good descriptive abilities. This was particularly true for the notched and centre hole specimens, which exhibited high accuracy. However, the predictive error of the shear specimen was large. This is because the stress state of the shear specimen is clearly different from that of NTR10 used for training, and the value is small, which induces the neural network model cannot accurately capture the state. However, the overall consistency for all specimens was satisfactory, considering that this prediction quality was achieved using only one specimen.

5.2. Comparison with experimental–numerical hybrid method

To further evaluate the machine learning assistance method, it was compared with the experimental–numerical hybrid method. The calibration process of the experimental–numerical hybrid method follows Ref. (Pütz et al., 2020), which was not described in this study. The calibration results are listed in Table 2, and the simulation results for each specimen are shown in Fig. 15 (green curve). Comparing the two methods, it can be observed that when using the experimental–numerical hybrid method for calibration, even if the shear specimen is directly involved in the calibration process, the prediction accuracy was slightly lower than that of other specimens; the effect of damage is slightly underestimated at low strain and slightly overestimated at high strain. Although the prediction error of the machine learning method for the shear specimen was slightly larger than that of the experimental–numerical hybrid method, the prediction performance of the other specimens performed well. In terms of prediction accuracy, the prediction accuracy of the machine learning assistance method is equally satisfactory; both methods can find a reliable parameter set. In terms of computing time, the machine learning assistance method replaced the iterative process of one particular calibration parameter (G_f) with a neural network, which reduced the time required for numerical simulation, the overall CPU time taken to create the database and training the model was 189.2 h. In the same computing environments, the overall CPU time for the experimental–numerical hybrid method was 296.7 h. Therefore, the machine learning assistance method has computational efficiency in comparison with the experimental–numerical hybrid method. In addition, the machine learning calibration strategy circumvents the process of extracting stress state variables from the numerical model of each calibrated specimen, and valuable information can be mined directly from the force–displacement response of only one specimen for calibration. The relative loading path during deformation is no longer a necessary condition for model calibration; this avoids the influence of critical element selection and time discretisation error on parameter calibration and further simplifies the calibration process, which is also a significant improvement.

6. Conclusion

In this study, the damage variables based on the damage initiation and fracture loci were coupled to a plastic model, and a semi-coupled fracture model was established. The parameters in the model were calibrated using an identification method based on machine learning assistance. The following conclusions were drawn:

(1) Considering the influence of hydrostatic stress and the third invariant of the stress deviator on the yield locus, the plastic yield model of the 6061 aluminium alloy was modified. The

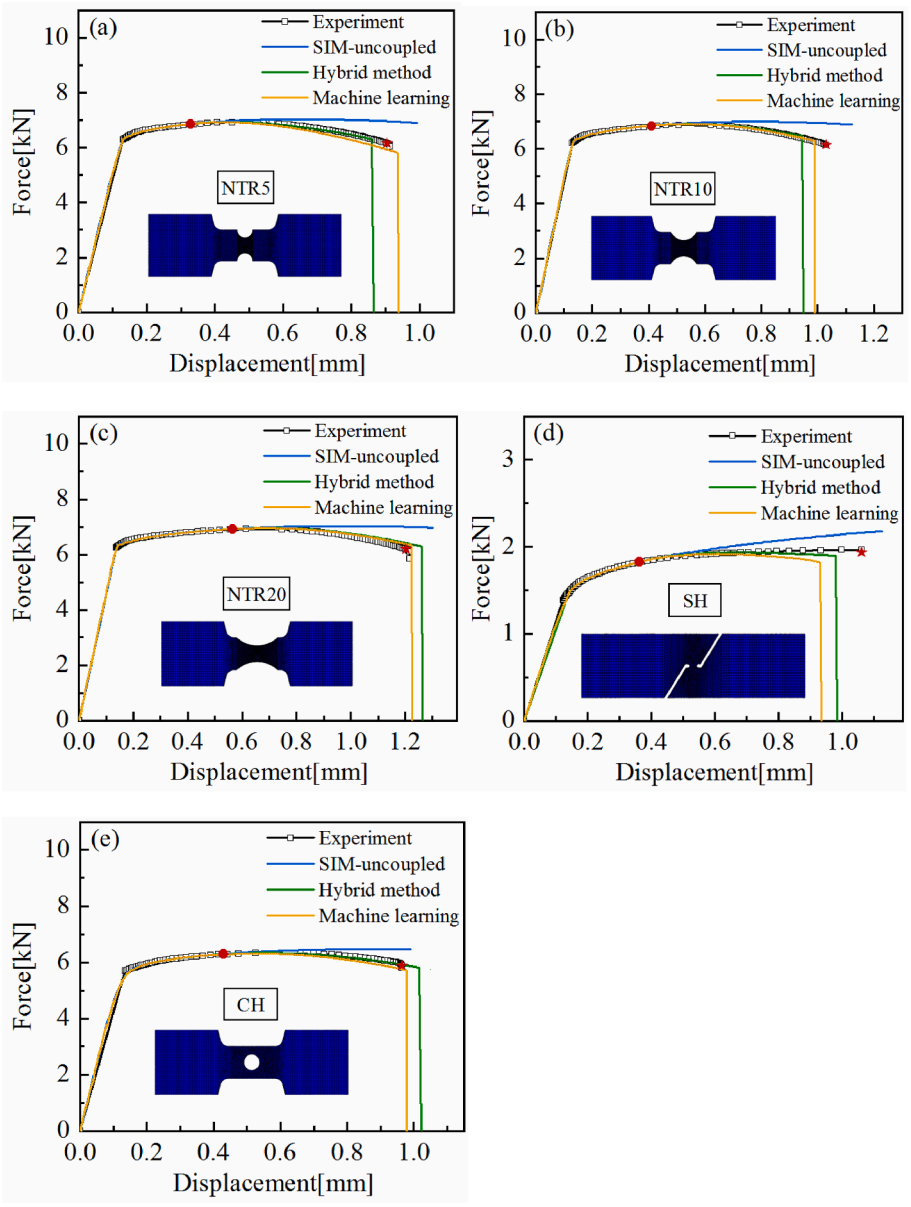


Fig. 15. Comparison of experimental and simulated force–displacement curves: a) NTR5 b) NTR10 c) NTR20 d) SH e) CH.

description ability of the yield model under various stress states was improved, which provides a basis for determining the damage initiation moment.

(2) Based on the uncoupled fracture criterion of Lou–Huh, the damage initiation indicator and fracture indicator of 6061 aluminium alloy were established, and a semi-coupled fracture model was constructed in combination with the damage evolution model. The results indicate that this semi-coupled fracture model can accurately predict the damage initiation, accumulation, and fracture of the 6061 aluminium alloy during forming.

(3) By substituting the extracted three-dimensional damage initiation locus into the integral formula of the damage initiation indicator, the parameters in the damage initiation model were determined based on the optimisation algorithm. Compared with the average stress state, the non-linearity of the loading paths is considered in this method.

(4) A parameter identification method based on machine learning assistance is proposed to replace the calibration process of the fracture and damage evolution models with a neural network. Compared with the traditional experimental–numerical hybrid method, this method circumvents the process of extracting the stress state evolution from the numerical models of all calibrated specimens, and a reliable parameter set can be extracted from the force–displacement response using only one specimen, which further simplifies the process of parameter calibration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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